Expt: 11a | Simulate static Routing Configuration using CISCO Packet Tracer

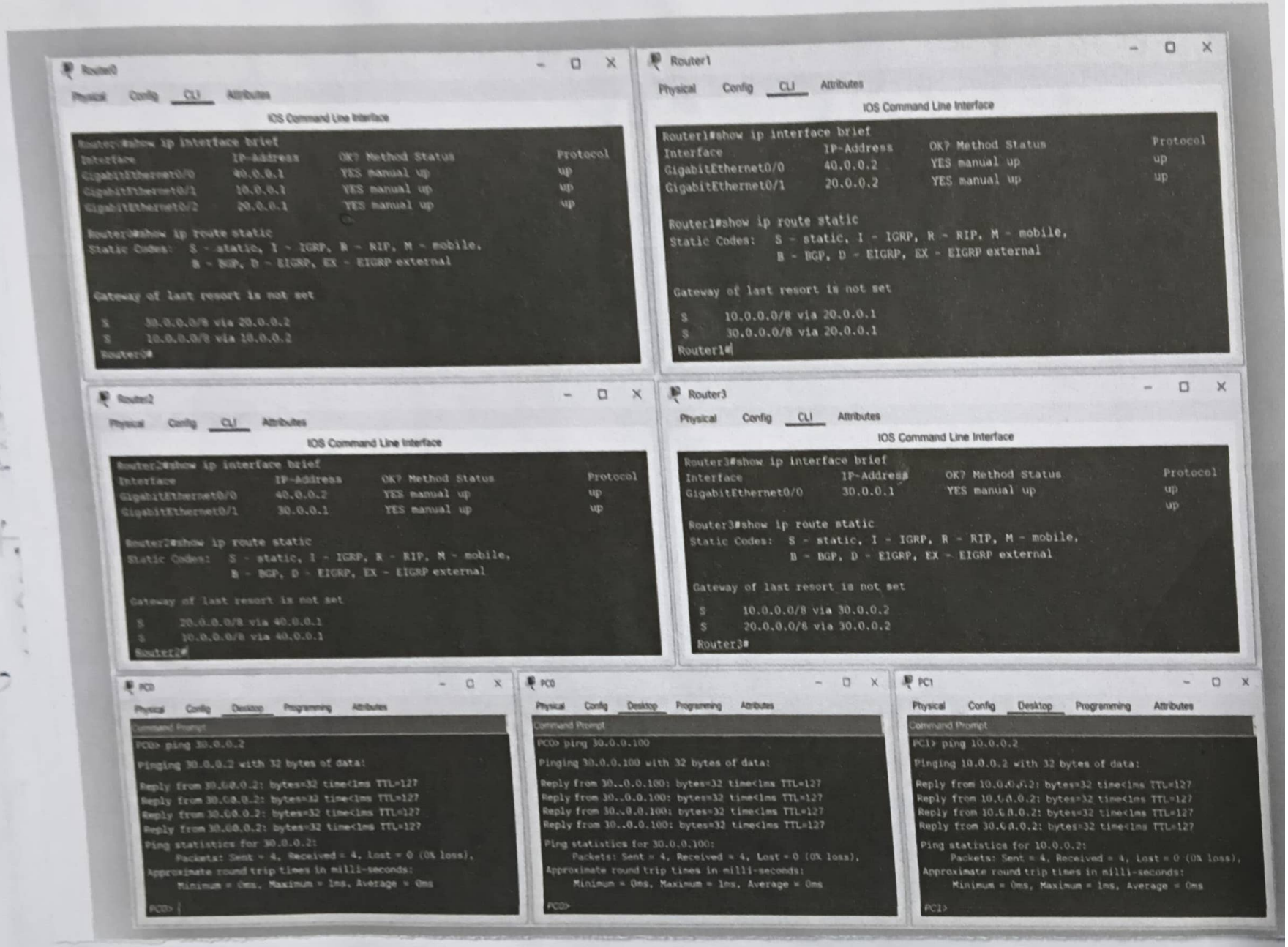
5.10.26

Aim:

To simulate a static Routing Configuration using CISO packet Tracer.

Procedure:

* Open Cisco Packet Tracer and create the given network Topology.
* Connect 3 routers and PCs using suitable cables
* Configure IP addresses for routers and PCs
* Enable all router interfaces using no shutdown
* Configure static routes on Router 0, Router1, and Router 2 using IP mobile
* Set main and backup routes using different Administrative Distance
* Test Connectivity
* Uses tracert to verify the routing path
* Disconnect, the main link and Verify that the backup route works
* Thus, a static routing is successfully configured and Verified


Router0
Physical Config CLI Attributes
IOS Command Line Interface
Router0#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 40.0.0.1 YES manual up up
GigabitEthernet0/1 10.0.0.1 YES manual up up
GigabitEthernet0/2 20.0.0.1 YES manual up up
Router0#show ip route static
Static Codes: S - static, I - IGRP, R - RIP, M - mobile,
B - BGP, D - EIGRP, EX - EIGRP external
Gateway of last resort is not set
S 30.0.0.0/8 via 20.0.0.2
S 10.0.0.0/8 via 10.0.0.2
Router0#
Router1
Physical Config CLI Attributes
IOS Command Line Interface
Router1#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 40.0.0.2 YES manual up up
GigabitEthernet0/1 20.0.0.2 YES manual up up
Router1#show ip route static
Static Codes: S - static, I - IGRP, R - RIP, M - mobile,
B - BGP, D - EIGRP, EX - EIGRP external
Gateway of last resort is not set
S 10.0.0.0/8 via 20.0.0.1
S 30.0.0.0/8 via 20.0.0.1
Router1#
Router2
Physical Config CLI Attributes
IOS Command Line Interface
Router2#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 40.0.0.2 YES manual up up
GigabitEthernet0/1 30.0.0.1 YES manual up up
Router2#show ip route static
Static Codes: S - static, I - IGRP, R - RIP, M - mobile,
B - BGP, D - EIGRP, EX - EIGRP external
Gateway of last resort is not set
S 20.0.0.0/8 via 40.0.0.1
S 10.0.0.0/8 via 40.0.0.1
Router2#
Router3
Physical Config CLI Attributes
IOS Command Line Interface
Router3#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 30.0.0.1 YES manual up up
Router3#show ip route static
Static Codes: S - static, I - IGRP, R - RIP, M - mobile,
B - BGP, D - EIGRP, EX - EIGRP external
Gateway of last resort is not set
S 10.0.0.0/8 via 30.0.0.2
S 20.0.0.0/8 via 30.0.0.2
Router3#
PC0
Physical Config Desktop Programming Attributes
Command Prompt
PC0> ping 30.0.0.2
Pinging 30.0.0.2 with 32 bytes of data:
Reply from 30.0.0.2: bytes=32 time<1ms TTL=127
Reply from 30.0.0.2: bytes=32 time<1ms TTL=127
Reply from 30.0.0.2: bytes=32 time<1ms TTL=127
Reply from 30.0.0.2: bytes=32 time<1ms TTL=127
Ping statistics for 30.0.0.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
PC0>
PC0
Physical Config Desktop Programming Attributes
Command Prompt
PC0> ping 30.0.0.100
Pinging 30.0.0.100 with 32 bytes of data:
Reply from 30.0.0.100: bytes=32 time<1ms TTL=127
Reply from 30.0.0.100: bytes=32 time<1ms TTL=127
Reply from 30.0.0.100: bytes=32 time<1ms TTL=127
Reply from 30.0.0.100: bytes=32 time<1ms TTL=127
Ping statistics for 30.0.0.100:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
PC0>
PC1
Physical Config Desktop Programming Attributes
Command Prompt
PC1> ping 10.0.0.2
Pinging 10.0.0.2 with 32 bytes of data:
Reply from 10.0.0.2: bytes=32 time<1ms TTL=127
Reply from 10.0.0.2: bytes=32 time<1ms TTL=127
Reply from 10.0.0.2: bytes=32 time<1ms TTL=127
Reply from 10.0.0.2: bytes=32 time<1ms TTL=127
Ping statistics for 10.0.0.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
PC1>